



UNIVERSITY OF THE PUNJAB

B.S. in Computer Science Third Semester : Fall 2023

Subject: Calculus and Analytical Geometry

Paper: MS-251 N

Roll No.
Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Define the following:

(6x5=30)

- I. Find domain and range of the function $y = \frac{x^2+x-2}{x^2-x-2}$?
- II. Evaluate the integral $\int \cos 2t \sin^5 2t \, dt$?
- III. Find parametric equations of the line through $(2, -1, 5)$ that is parallel to $\langle -1, 2, 7 \rangle$.
- IV. Find second order derivative of $f(x) = (1 + 3x)^8$?
- V. Show that $y = \cos x$ and $y = \sin x$ are the solutions to $y'' + y = 0$.
- VI. Find an equation of the plane through the points $P_1(1, 2, -1)$, $P_2(2, 3, 1)$, and $P_3(3, -1, 2)$.

Q.2. Answer the following questions.

(3x10=30)

- I. Find the intervals on which $g(x) = -x^3 + 12x + 5$, $-3 \leq x \leq 3$ is increasing and decreasing. Where does the function assume the extreme values and what are these values?
- II. a) Evaluate the integral $\int r^4 \left(7 - \frac{r^6}{10}\right)^3 dr$?
- b) Evaluate the integral $\int_0^{\pi/2} \frac{1}{2} (\cos x + |\cos x|) dx$?
- III. Find parametric equations for the line in which the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$ intersect?

Here's how to find the domain and range of the function $Y = \frac{x^2 + x - 2}{x^2 - x - 2}$:

1. Find the Domain:

The domain of a function is the set of all possible input values (x-values) for which the function is defined. Rational functions (functions that are fractions of polynomials) are undefined when the denominator is equal to zero. Therefore, we need to find the values of x that make the denominator zero.

We set the denominator equal to zero and solve for x:

$$x^2 - x - 2 = 0$$

This quadratic equation can be factored as:

$$(x - 2)(x + 1) = 0$$

This gives us two solutions: $x = 2$ and $x = -1$.

These values make the denominator zero, so they are excluded from the domain. The domain is all real numbers except $x = 2$ and $x = -1$. In interval notation, the domain is:

$$(-\infty, -1) \cup (-1, 2) \cup (2, \infty)$$

2. Find the Range:

Finding the range of a rational function can be more challenging. Let's simplify the function first by factoring the numerator and denominator:

$$y = \frac{(x+2)(x-1)}{(x-2)(x+1)}$$

Notice that there is a common factor of $(x - 1)$ in the numerator and denominator. However, we cannot cancel them out because that would change the function's behavior at 1.

To find the range, we can analyze the behavior of the function as x approaches its excluded values and as x approaches positive and negative infinity. We can also look for horizontal asymptotes.

As x approaches ∞ or $-\infty$, the function approaches 1 (the ratio of the leading coefficients of the numerator and denominator). Therefore, there is a horizontal asymptote at $y = 1$.

However, since the numerator and denominator have different roots, there will be a vertical asymptote at $x = -1$ and $x = 2$. The function will approach $\pm\infty$ near these asymptotes. Let's consider the value of the function at 1:

$$y = \frac{1^2 + 1 - 2}{1^2 - 1 - 2} = \frac{0}{-2} = 0$$

The function is 0 at $x = 1$. Since there is a horizontal asymptote at $y = 1$, the range will include all real numbers except possibly $y = 1$.

To confirm if $y = 1$ is in the range, we would need to solve the equation:

$$1 = \frac{x^2 + x - 2}{x^2 - x - 2}$$

$$x^2 - x - 2 = x^2 + x - 2$$

Since 0 is in the domain, $y = 0$ is in the range.

Therefore, the range is all real numbers. In interval notation, the range is $(-\infty, \infty)$.

The domain of the function is

$(-\infty, -1) \cup (-1, 2) \cup (2, \infty)$, and the range of the function is $(-\infty, \infty)$.

Here's how to find the equation of the plane passing through the points $P_1(1, 2, -1)$, $P_2(2, 3, 1)$, and $P_3(3, -1, 2)$.

1. Find two vectors in the plane:

We can find two vectors lying in the plane by subtracting the coordinates of the points. Let's use:

$$\vec{v} = P_2 - P_1 = (2 - 1, 3 - 2, 1 - (-1)) = (1, 1, 2)$$

$$\vec{w} = P_3 - P_1 = (3 - 1, -1 - 2, 2 - (-1)) = (2, -3, 3)$$

2. Find the normal vector:

The normal vector $\mathbf{n}$ to the plane is given by the cross product of $\vec{v}$ and $\vec{w}$:

$$\mathbf{n} = \vec{v} \times \vec{w} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 2 \\ 2 & -3 & 3 \end{vmatrix} = \hat{i}(3 - (-6)) - \hat{j}(3 - 4) + \hat{k}(-3 - 2)$$

3. Find the equation of the plane:

The equation of a plane is given by:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

where (A, B, C) is the normal vector and (x_0, y_0, z_0) is a point on the plane. Using $P_1(1, 2, -1)$ and $\mathbf{n}(9, 1, -5)$, we get:

$$9(x - 1) + 1(y - 2) - 5(z - (-1)) = 0$$

$$9m - 9 + y - 2 - 5z - 5 = 0$$

$$-5z - 16 = 0$$

Therefore, the equation of the plane is

$$-5z = 16.$$

Part 1: Showing $y = \cos x$ and $y = \sin x$ are solutions to $y'' + y = 0$

To show that $y = \cos x$ and $y = \sin x$ are solutions, we need to find their second derivatives and substitute them into the given differential equation.

1. For $y = \cos x$:

○ The first derivative is $y' = -\sin x$.

○ The second derivative is $y'' = -\cos x$.

○ Substituting into the equation:

$$y'' + y = -\cos x + \cos x = 0$$

. Therefore, $y = \cos x$ is a solution.

2. For $y = \sin x$:

○ The first derivative is $y' = \cos x$.

0 The second derivative is
 $= -\sin c.$

0 Substituting into the
equation:

$y'' + y = -\sin x +$
 $\sin a.$ Therefore, $y \sin c$
is a
solution.



3. Find the second-order derivative of

$$f(x) = (1 + 3x)^8$$

First, we find the first derivative using the chain rule:

$$f'(x) = 8(1 + 3x)^7 \cdot 3 = 24(1 + 3x)^7$$

Next we find the second derivative, again using the chain rule:

$$f''(x) = 24 \cdot (1 + 3x)^6 \cdot 3 = 504(1 + 3x)^6$$

Therefore, the second-order derivative of

$$f(x) = (1 + 3x)^8 \text{ is } 504(1 + 3x)^6.$$

Final Answer: The final answer is $504(1 + 3r)^6$



2. Find parametric equations of the line through $(2, -1, 5)$ that is parallel to $(-1, 2, 7)$.

The parametric equations of a line passing through a point (x_0, y_0, z_0) and parallel to a vector $\mathbf{v} = (a, b, c)$ are given by:

$x = x_0 + at$, $y = y_0 + bt$, $z = z_0 + ct$ where t is a parameter.

In this case, the point is $(2, -1, 5)$, so $(x_0, y_0, z_0) = (2, -1, 5)$, and the direction vector is $\mathbf{v} = (-1, 2, 7)$, so $(a, b, c) = (-1, 2, 7)$.

Therefore, the parametric equations of the line are:

$$x = 2 - t$$
$$y = -1 + 2t$$

$$z = 5 + 7t$$

The parametric equations of the line are $x = 2 - t$, $y = -1 + 2t$, and $z = 5 + 7t$.

Here's how to solve these problems:

1. Evaluate the integral $\int \cos 2t \sin^5 2t \, dt$

We can use substitution to solve this integral. Let $u = \sin 2t$. Then, the derivative of u with respect to t is:

$$du = 2 \cos 2t \, dt$$

So, $\cos 2t \, dt = \frac{1}{2} du$. Substituting into the integral, we get:

$$\int \cos 2t \sin^5 2t \, dt = \frac{1}{2} \int u^5 \, du$$

Now, we can integrate u^5 :

$$\frac{1}{2} \int u^5 \, du = \frac{1}{2} \cdot \frac{u^6}{6} + C = \frac{u^6}{12} + C$$

Finally, substitute back $u = \sin 2t$:

$$\frac{\sin^6 2t}{12} + C$$

Therefore, the integral evaluates to $\frac{\sin^6 2t}{12} + C$.

to solve this problem step-by-step:

1. Find the derivative:

To determine where the function is increasing or decreasing, we need to find its derivative, $g'(x)$. The derivative of $g(x) = -x^3 + 12x + 5$ is:

$$g'(x) = -3x^2 + 12$$

2. Find critical points:

Critical points occur where the derivative is zero or undefined. In this case, the derivative is a polynomial, so it's defined everywhere. We set the derivative equal to zero and solve for x :

Here's how

$$-3x + 12 = 0$$

$$3x = 12$$

So, the critical points are $x = 2$ and $x = -2$.

3. Determine intervals of increase and decrease:

Now we test the intervals determined by the critical points $(-3, -2)$, $(-2, 2)$, and $(2, 3)$ to see if the derivative is positive (increasing) or negative (decreasing).

- Interval $(-3, -2)$: Choose a test point, say $x = -2.5$. Then $g'(-2.5) = -3(-2.5)^2 + 12 = -3(6.25) = -18.75$. Therefore, $g(x)$ is decreasing on the interval $(-3, -2)$.
- Interval $(-2, 2)$: Choose a test point, say $x = 0$. Then $g'(0) = -3(0)^2 + 12 = 12 > 0$. Therefore, $g(x)$ is increasing on the interval $(-2, 2)$.
- Interval $(2, 3)$: Choose a test point, say $x = 2.5$. Then $g'(2.5) = -3(2.5)^2 + 12 = -3(6.25) = -18.75$. Therefore, $g(x)$ is decreasing on the interval $(2, 3)$.

4. Find extreme values:

Since the function changes from decreasing to increasing at $x = -2$, there is a local minimum at $x = -2$. The value of the function at this point is:

$$g(-2) = (-2)^3 + 12(-2) + 5 = -8 - 24 + 5 = -11$$

Since the function changes from increasing to decreasing at $x = 2$, there is a local maximum at $x = 2$. The value of the function at this point is:

$$g(2) = (2)^3 + 12(2) + 5 = 8 + 24 + 5 = 21$$

We also need to check the endpoints of the interval

$$g(-3) = (-3)^3 + 12(-3) + 5 = -27 - 36 + 5 = -44$$

$$g(3) = (3)^3 + 12(3) + 5 = 27 + 36 + 5 = 44$$

Comparing the values at the critical points and endpoints, the absolute minimum is -11 at $x = -2$, and the absolute maximum is 21 at $x = 2$.

Therefore, $g(x)$ is increasing on the interval $(-2, 2)$ and decreasing on the intervals $(-3, -2)$ and $(2, 3)$. The extreme values are a local minimum of -11 at $x = -2$ and a local maximum of 21 at $x = 2$. The absolute minimum is -11 and the absolute maximum is 21 .

Here's how

to solve these integral problems:

a) Evaluate the integral $\int r^4 \left(7 - \frac{r^5}{10}\right) dr$

This integral requires a substitution. Let $u = 7 - \frac{r^5}{10}$.

Then, the derivative of u with respect to r is:

$$\frac{du}{dr} = -\frac{5r^4}{10} = -\frac{r^4}{2}$$

We can rearrange this to get:

$$-2du = r^4 dr$$

Now substitute u and $-2du$ into the integral:

$$\int r^4 \left(7 - \frac{r^5}{10}\right) dr = \int u^3 (-2du) = -2 \int u^3 du$$

Now integrate with respect to u :

$$-2 \int u^3 du = -2 \left(\frac{u^4}{4} \right) + C = -\frac{u^4}{2} + C$$

Finally, substitute back for u :

$$-\frac{1}{2} \left(7 - \frac{r^5}{10}\right)^4 + C$$

Therefore, the solution to the integral is

$$-\frac{1}{2} \left(7 - \frac{r^5}{10}\right)^4 + C$$

b) Evaluate the integral $\int_0^{\pi} (\cos x + |\cos x|) dx$

The absolute value of $\cos$ is defined as:

$$|x| = \begin{cases} \cos x & \text{if } \cos x \geq 0 \\ -\cos x & \text{if } \cos x < 0 \end{cases}$$

x

In the interval $[0, \pi]$, $\cos$ is non-negative from 0 to $\frac{\pi}{2}$ and non-positive from $\frac{\pi}{2}$ to π . Therefore, we can split the integral into two parts:

$$\int_0^{\pi} (\cos x + |\cos x|) dx = \frac{1}{2} \left[\int_0^{\pi/2} (\cos x + \cos x) dx + \int_{\pi/2}^{\pi} (\cos x - \cos x) dx \right]$$

$$= \frac{1}{2} \left[\int_0^{\pi/2} 2 \cos x dx - \int_{\pi/2}^{\pi} 0 dx \right] = \int_0^{\pi/2} \cos x dx$$

Here's how

Now integrate:

$\cos x dx [\sin(-) - \sin(0)] = 1$ — Therefore, the solution to the integral is 1.

to find the parametric equations for the line of intersection of the two planes:

1. Find a point on the line:

To find a point on the line of intersection, we can set one of the variables to a convenient value and solve for the other two. Let's set $z = 0$ in both plane equations:

$$\text{Plane 1: } 3x - 6y - 2(0) = 15 \Rightarrow 3x - 6y = 15$$

$$\text{Plane 2: } 2x + y - 2(0) = 5 \Rightarrow 2x + y = 5$$

Now we have a system of two linear equations with two unknowns:

$$3x - 6y = 15$$

$$2x + y = 5$$

We can solve this system using substitution or elimination. Let's use elimination. Multiply the second equation by 6 to eliminate Y :

$$3x - 6y = 15$$

$$12x + 6y = 30$$

Add the two equations:

$$15x = 45 \Rightarrow x = 3$$

Substitute 3 into the second equation:

$$2(3) + y = 5 \Rightarrow y = -1$$

So, we have a point on the line of intersection:

$$(3, -1, 0).$$

2. Find the direction vector:

The direction vector of the line of intersection is perpendicular to the normal vectors of both planes. The normal vectors are:

$$\begin{aligned} \text{Plane 1: } \mathbf{n}_1 &= \begin{pmatrix} 3 \\ -6 \\ -2 \end{pmatrix} \\ \text{Plane 2: } \mathbf{n}_2 &= \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \end{aligned}$$

112

The direction vector $\mathbf{v}$ is given by the cross product of the normal vectors:

$$\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{pmatrix} 3 \\ -6 \\ -2 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} (-6)(-2) - (-2)(1) \\ (-2)(2) - (3)(-2) \\ (3)(1) - (-6)(2) \end{pmatrix}$$

3. Write the parametric equations:

Here's how

Now we have a point $(3, -1, 0)$ and a direction

$$\begin{pmatrix} 14 \\ 2 \\ 15 \end{pmatrix}$$

vector. The parametric equations for the line are:

$$x = 3 + 14t$$

$$y = -1 + 2t$$

$$z = 15t$$

The parametric equations for the line in which the planes intersect are $x = 3 + 14t$, $y = -1 + 2t$, and $z = 15t$.

